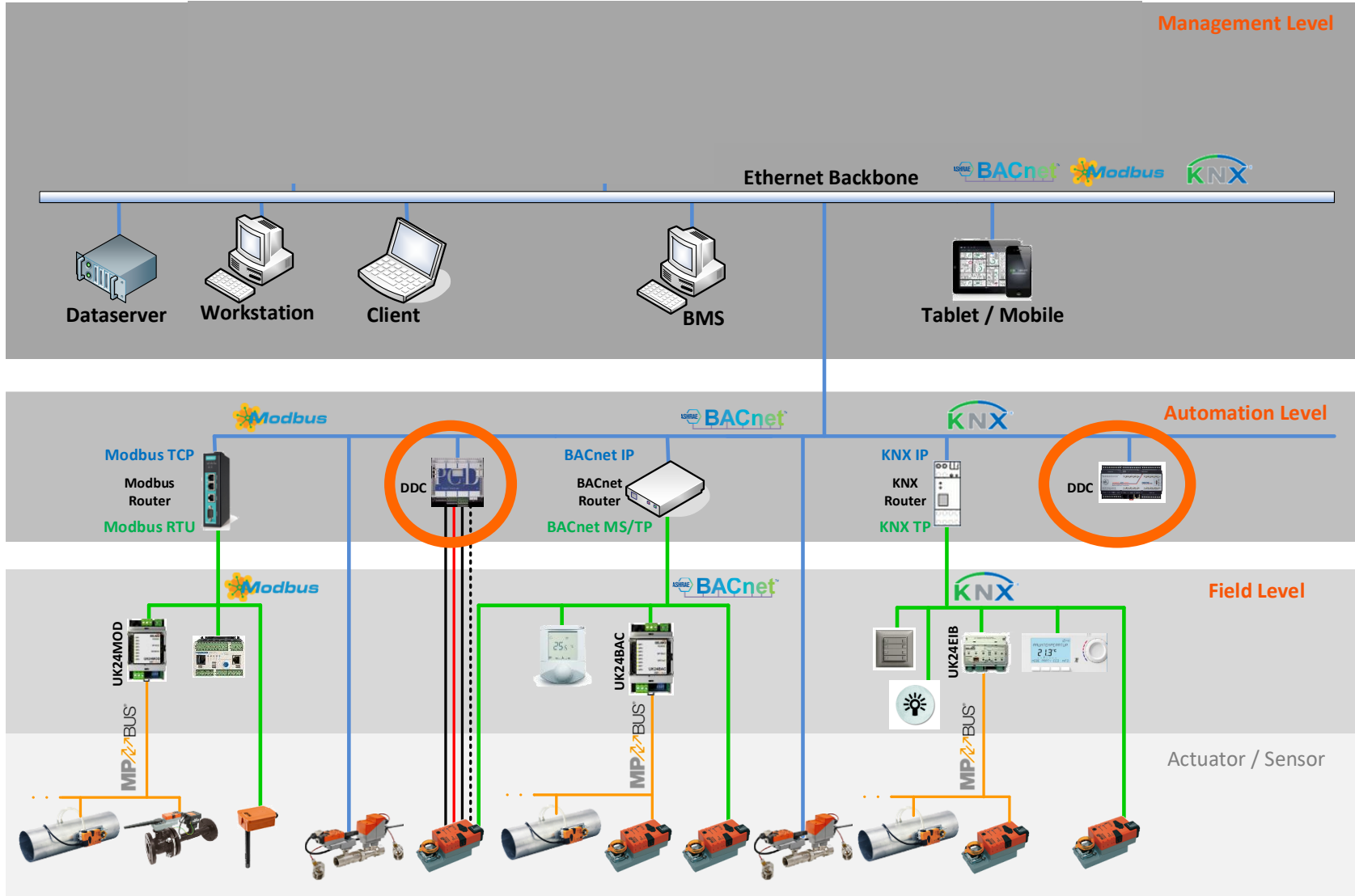


DDC Integration

March 4th – 5th, 2026

Building Topology - DDC

TODAY



What does every bus protocol need?

- Defined transmitters and receivers:

① Devices / Nodes



- Defined transmission path:



② Wiring and topology

- Unique addresses:

③ Addressing



- Defined data representation / packaging:

④ Data points, operating values



- Defined rules for transmission:

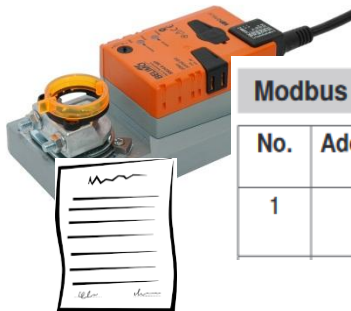
⑤ communication principles



The DDC needs to know about:

Defined data representation / packaging:

④ Data points, operating values

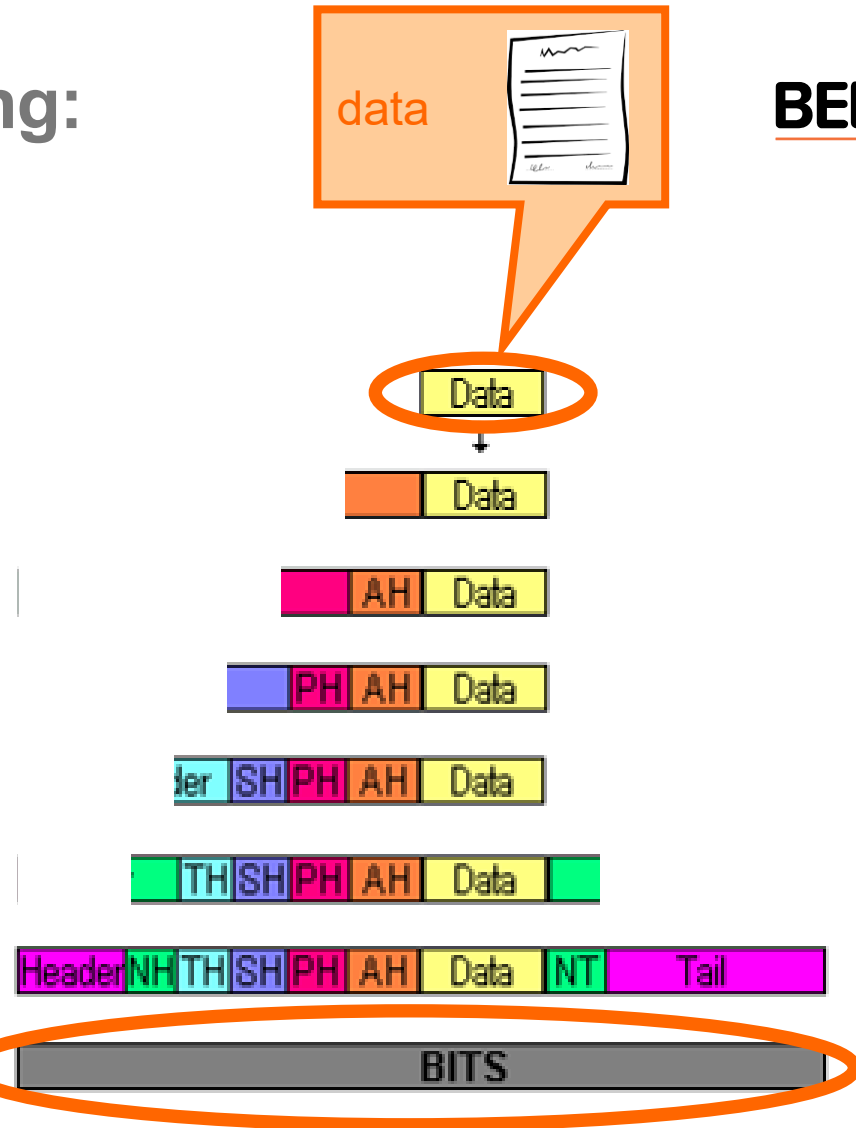
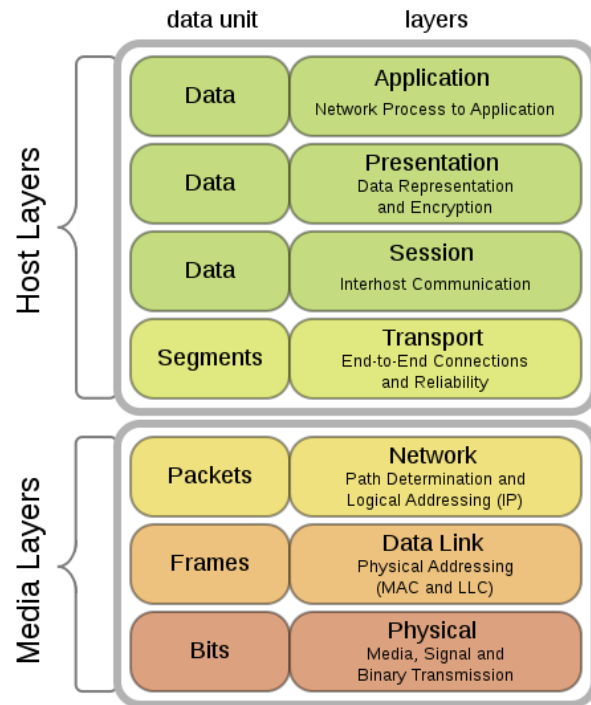


Modbus Register Description

No.	Address	Description Comment	Range Enumeration	Unit	Scaling	Access
1	0	Setpoint Setpoint for actuator between Min (Register No. 106) and Max (No. 107)	0...10000 Default: 0	%	0.01	R / W

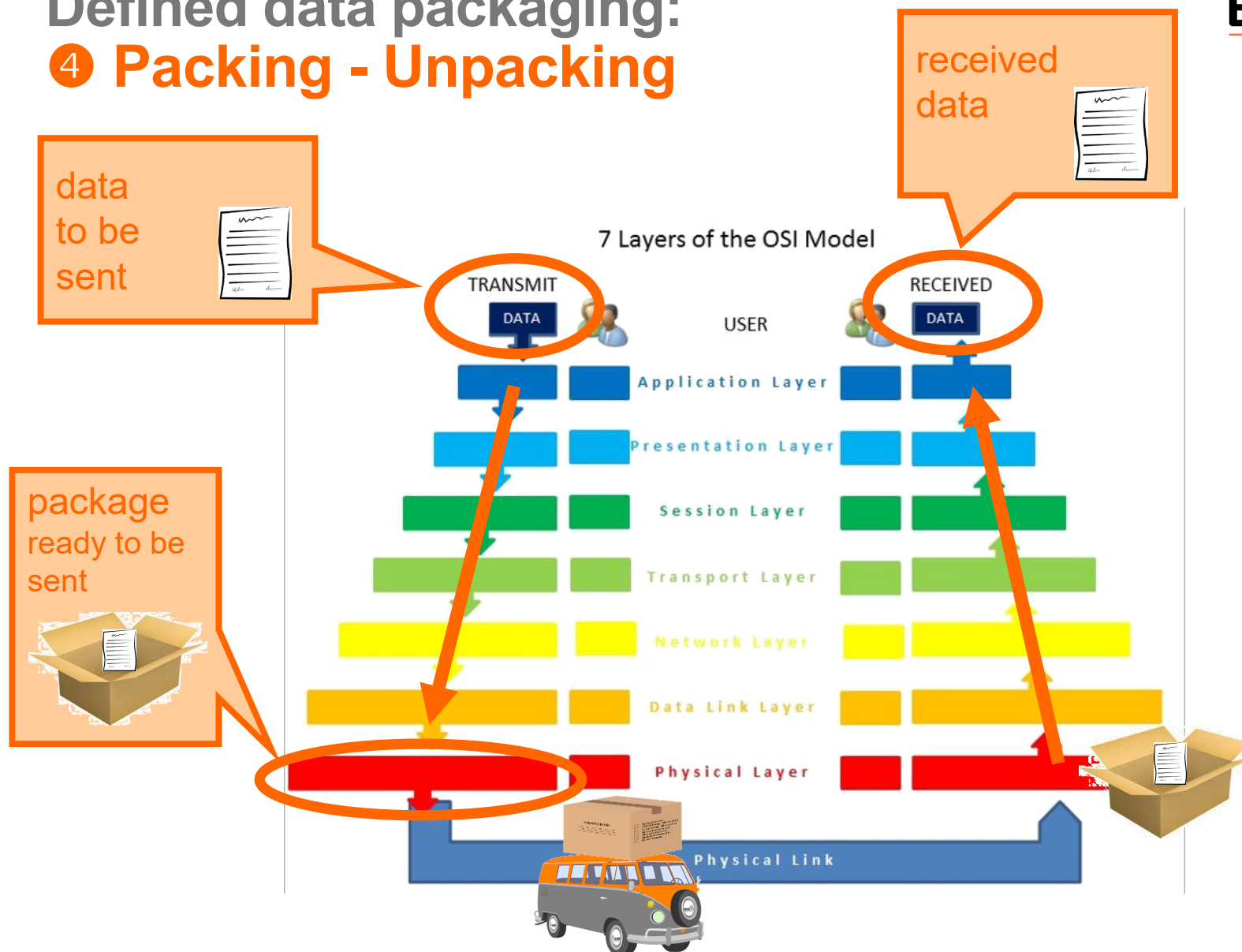
Defined data packaging:

④ OSI Model Layers



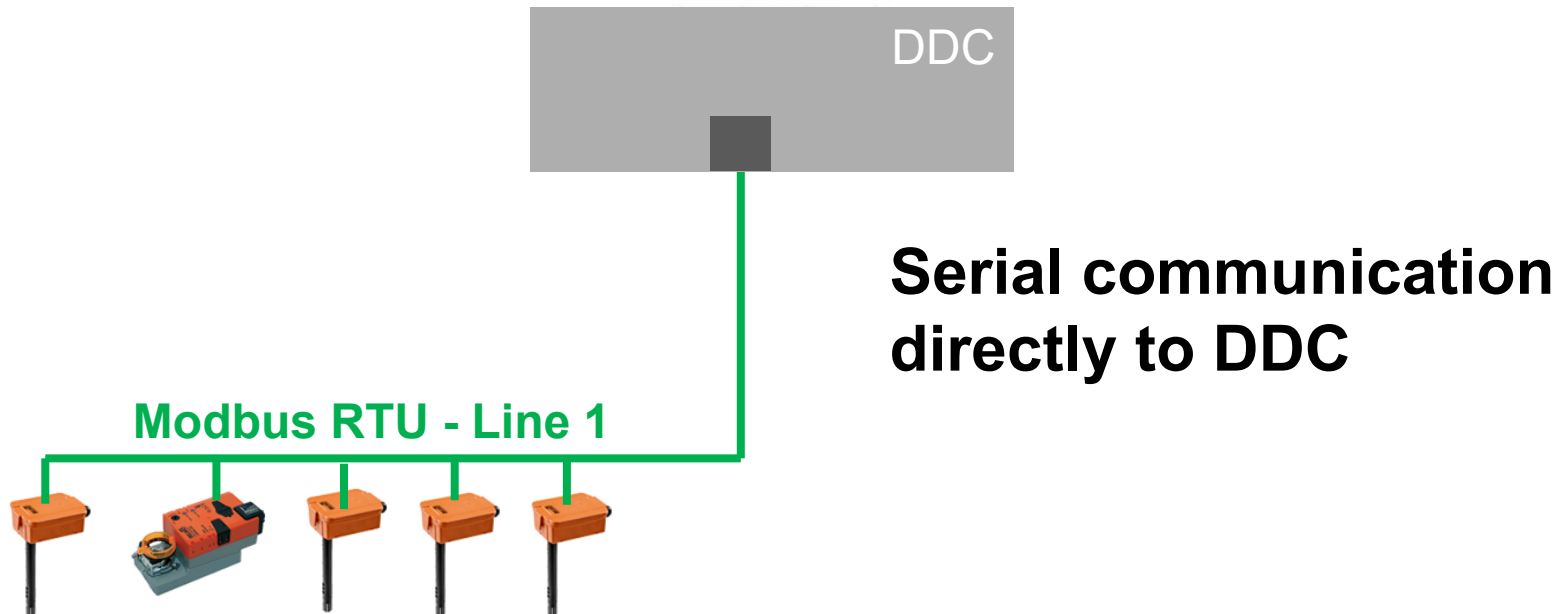
Defined data packaging:

④ Packing - Unpacking

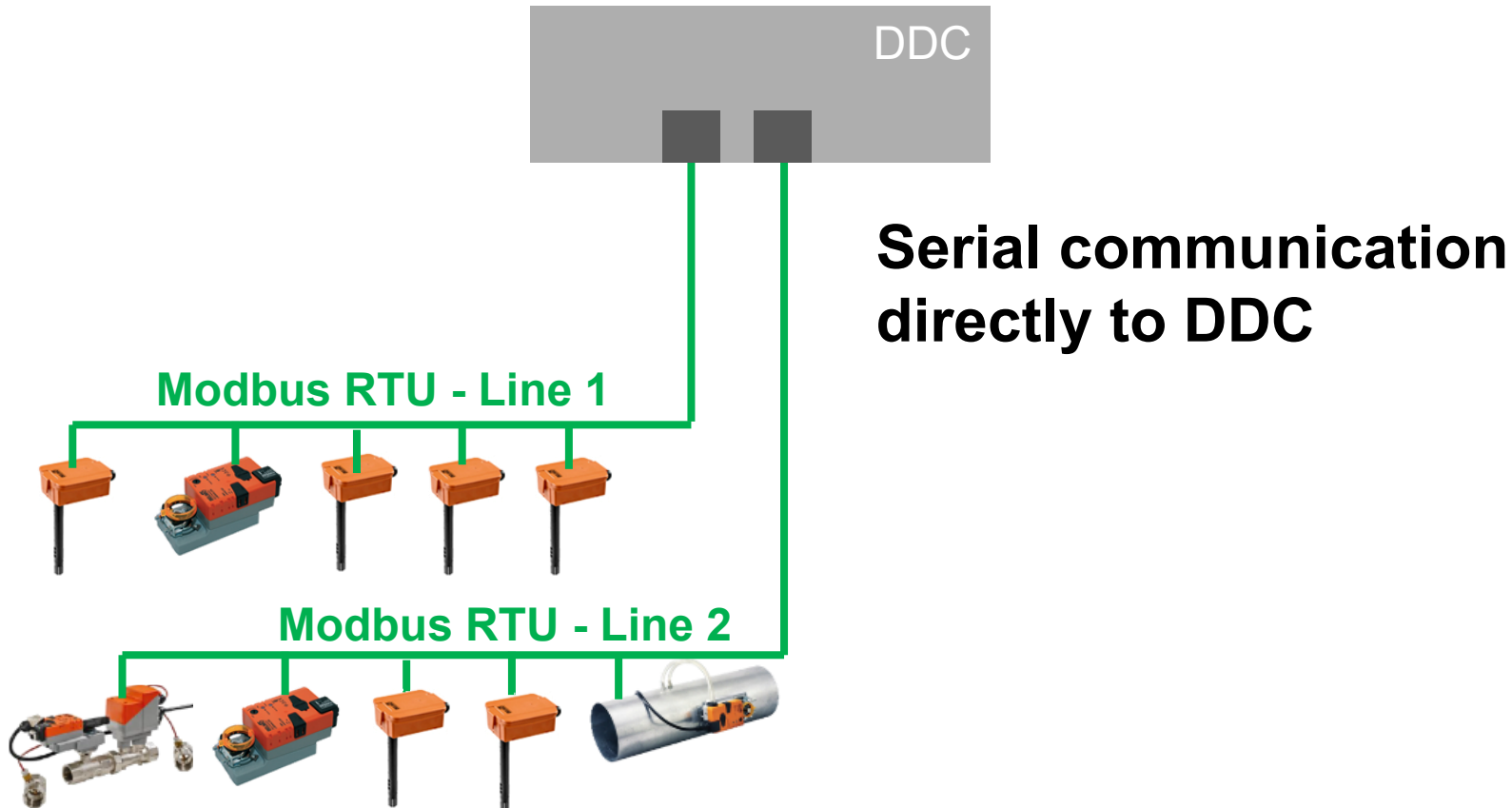


Integration of Field Communication in DDC

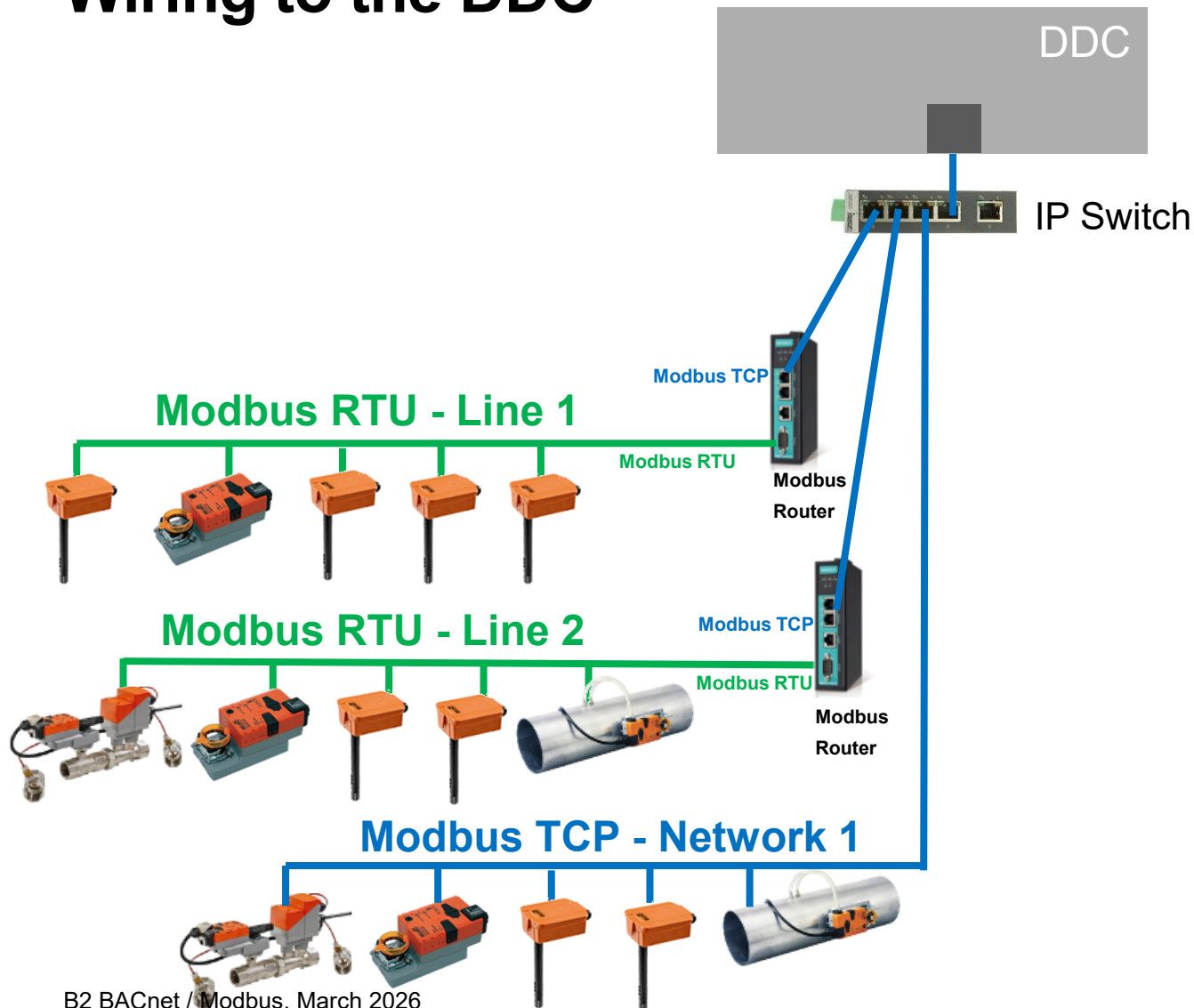
Wiring to the DDC



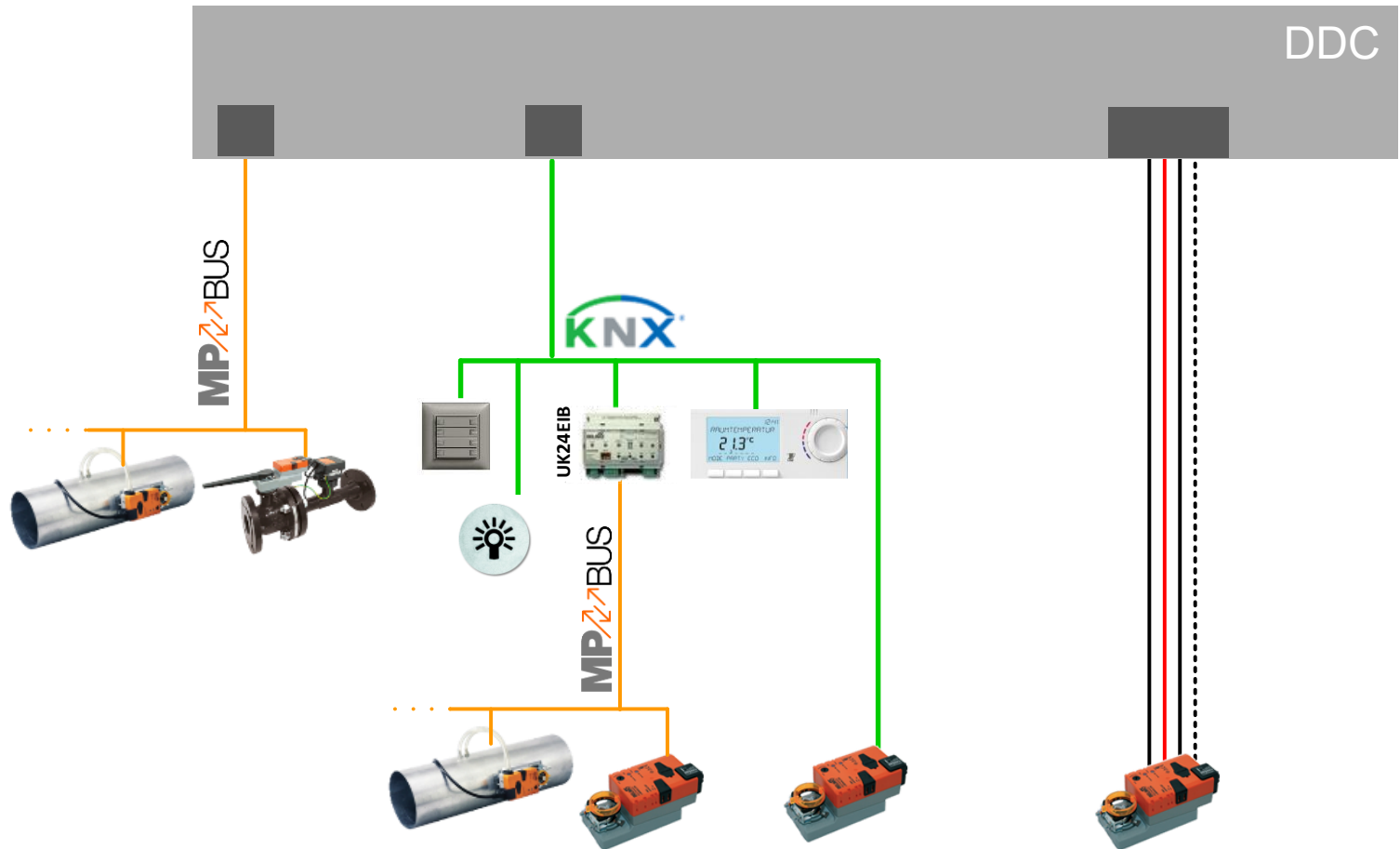
Integration of Field Communication in DDC Wiring to the DDC



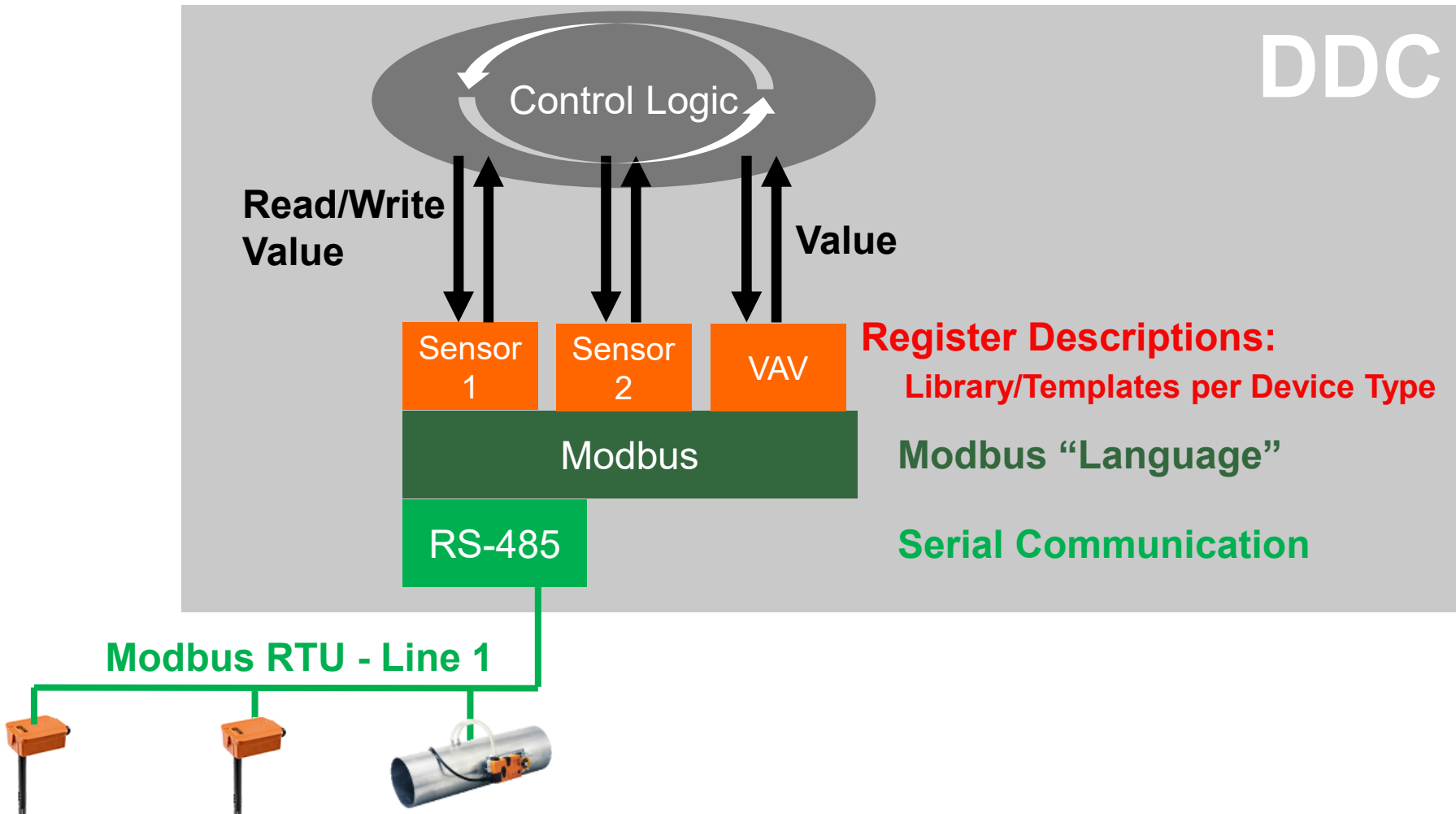
Integration of Field Communication in DDC Wiring to the DDC



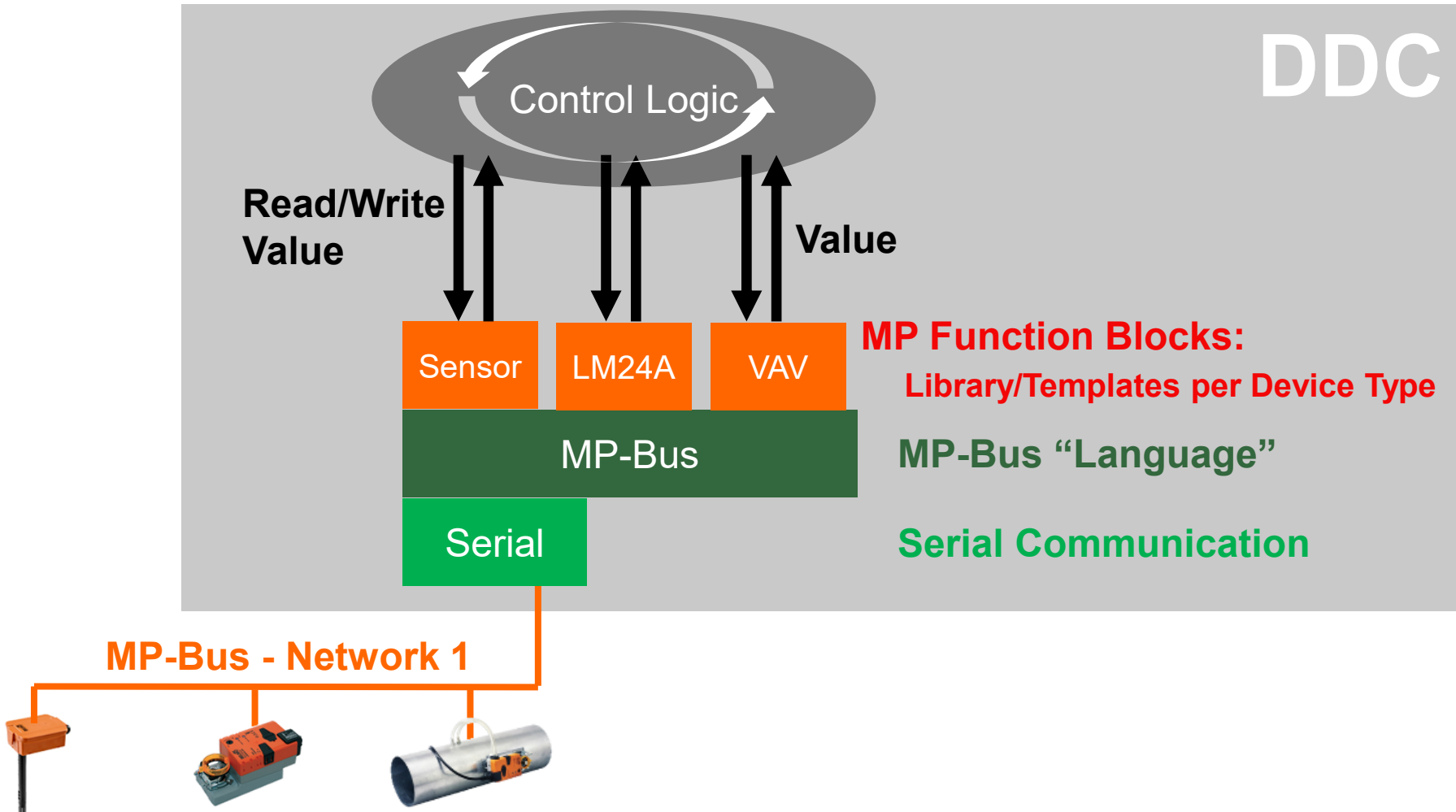
Integration of Field Communication in DDC Wiring to the DDC



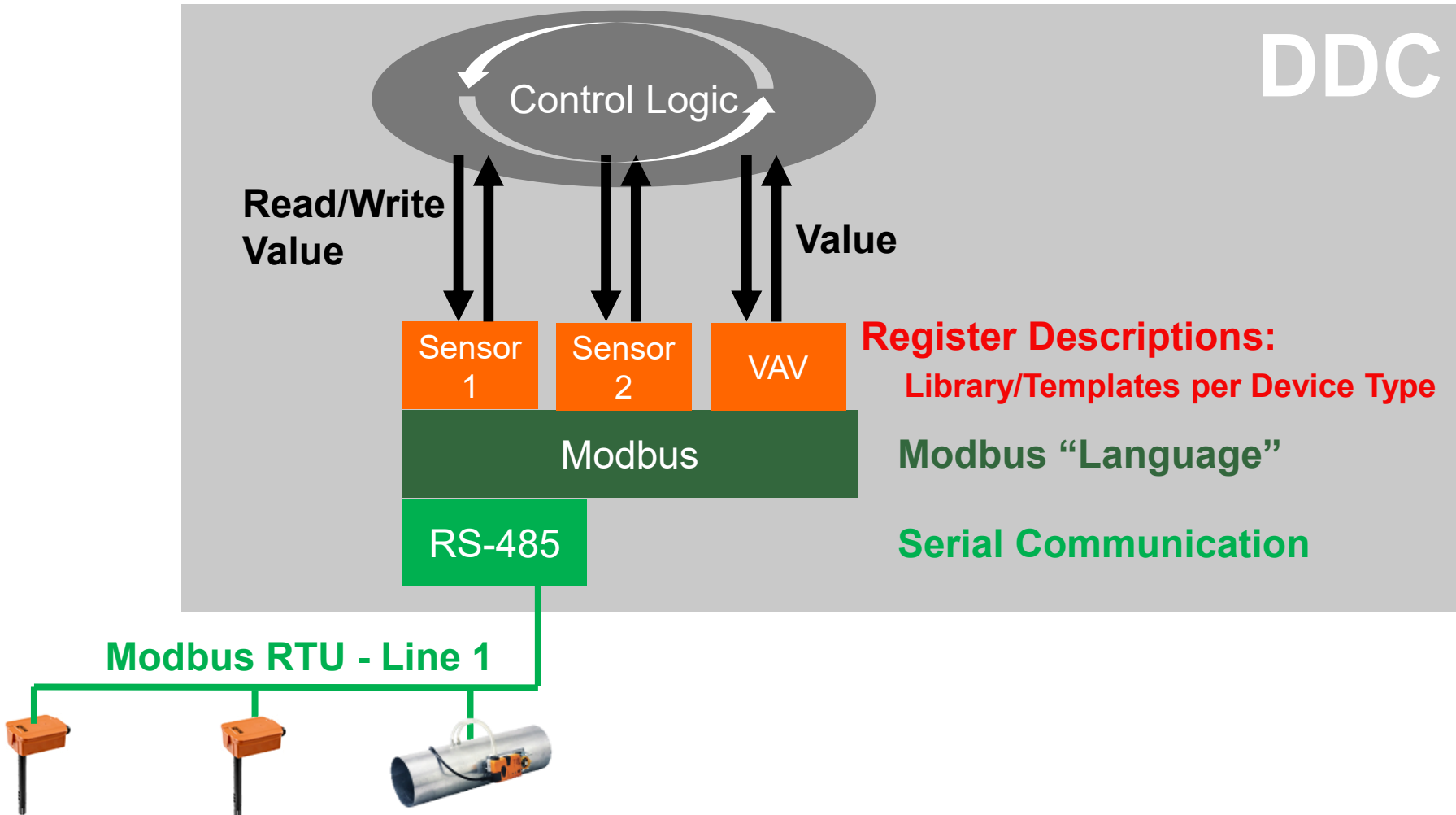
Integration of Field Communication in DDC Wiring to the DDC



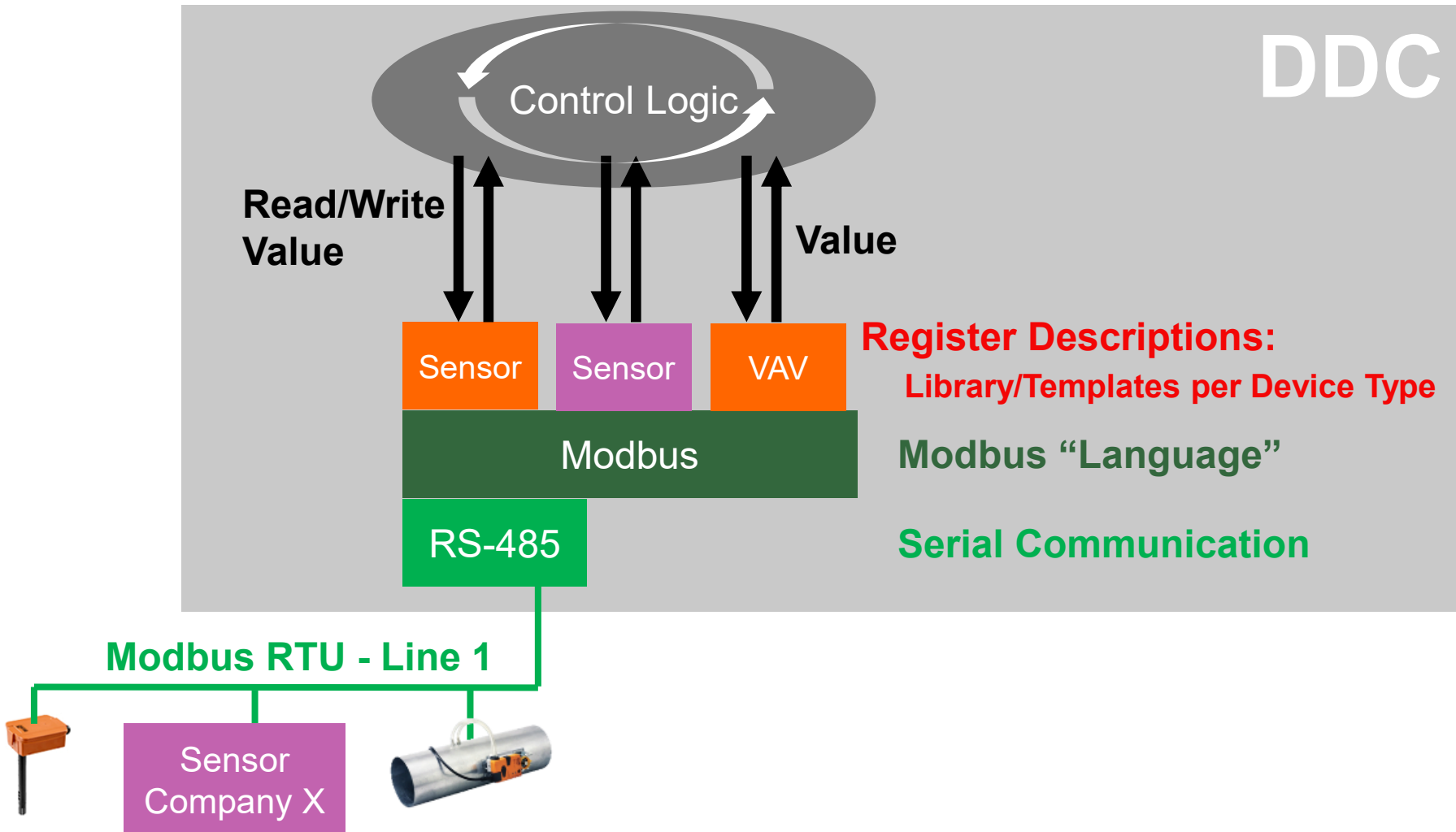
Integration of Field Communication in DDC Wiring to the DDC



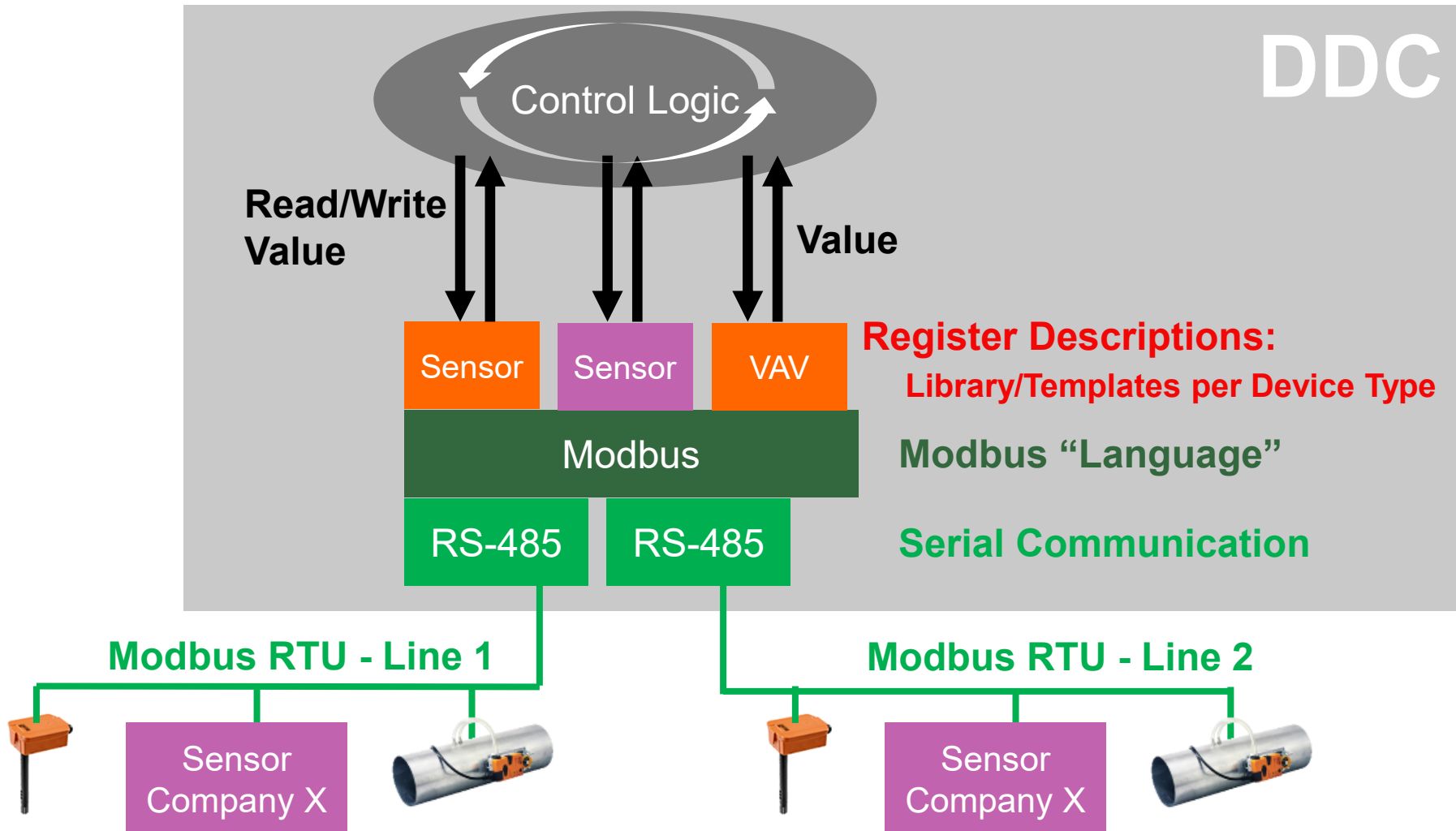
Integration of Field Communication in DDC Wiring to the DDC



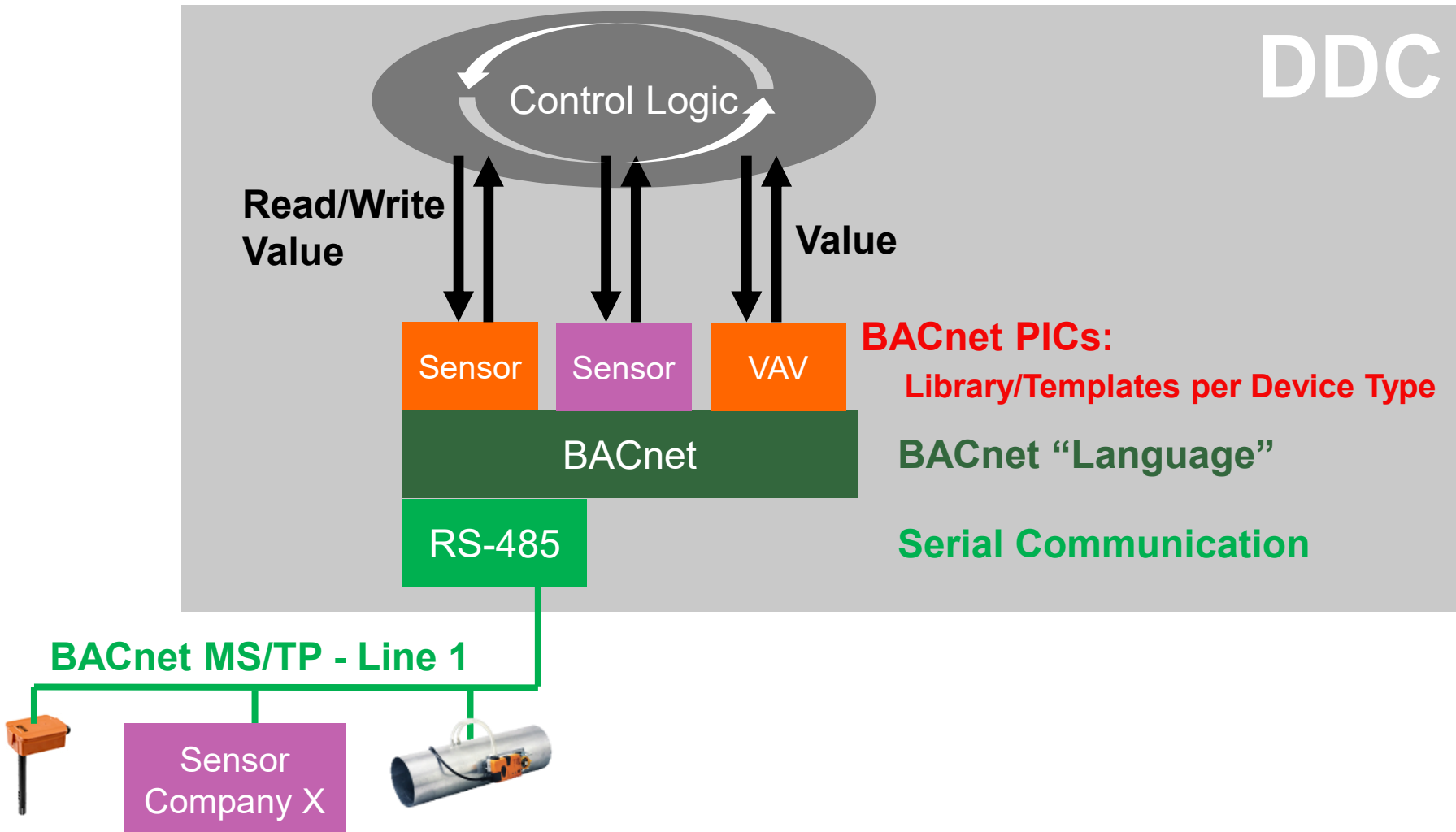
Integration of Field Communication in DDC Wiring to the DDC



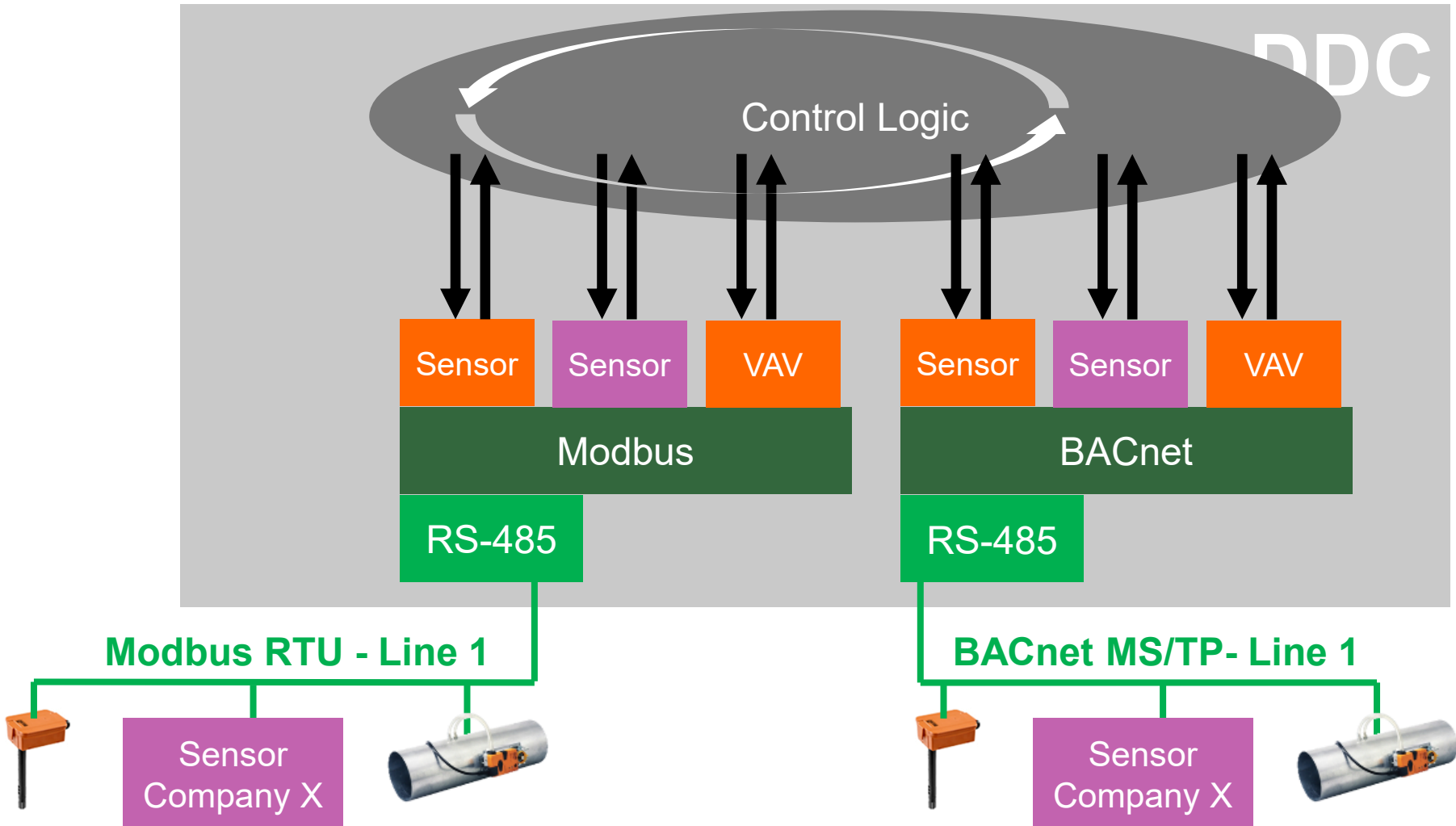
Integration of Field Communication in DDC Wiring to the DDC



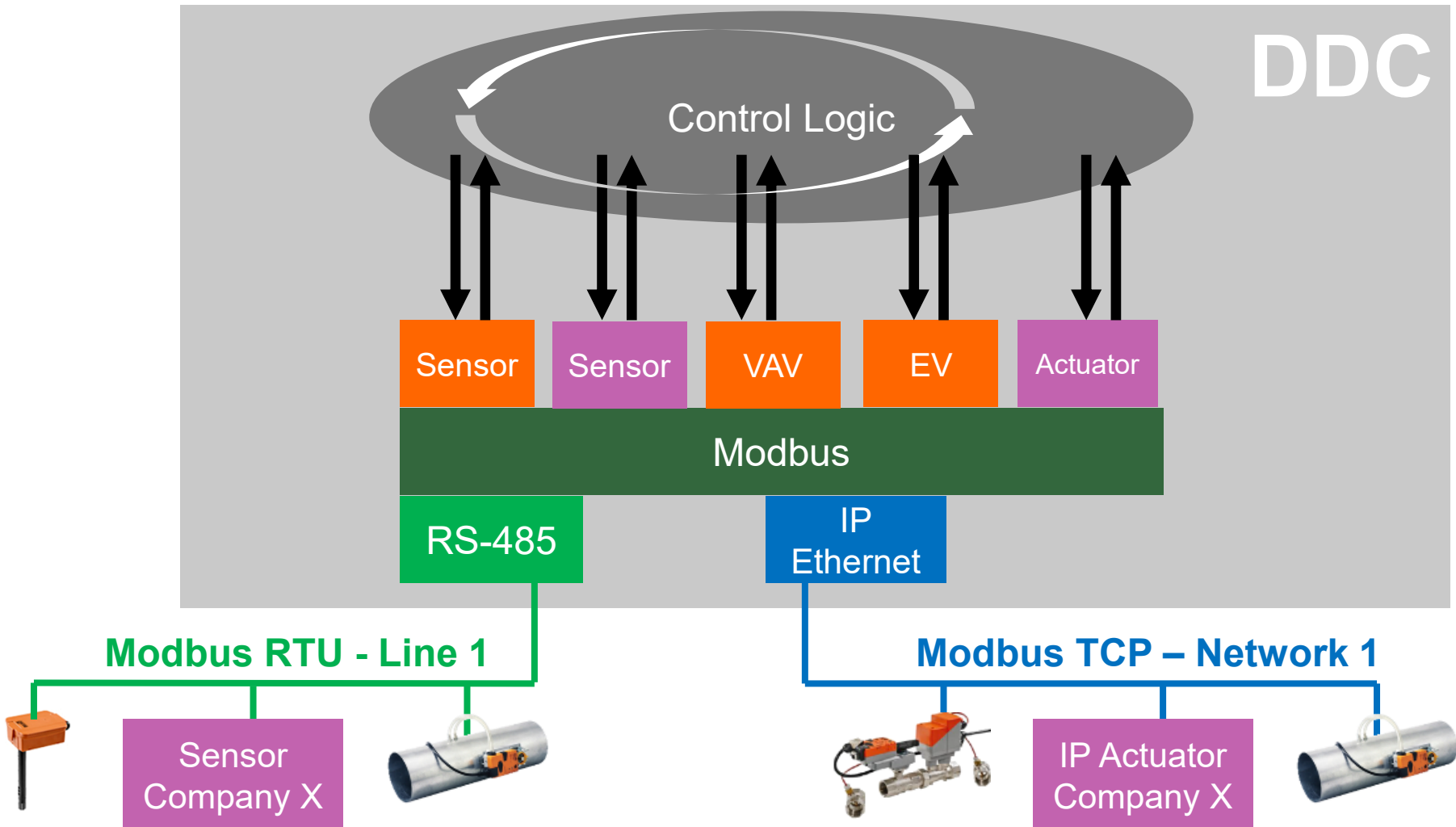
Integration of Field Communication in DDC Wiring to the DDC

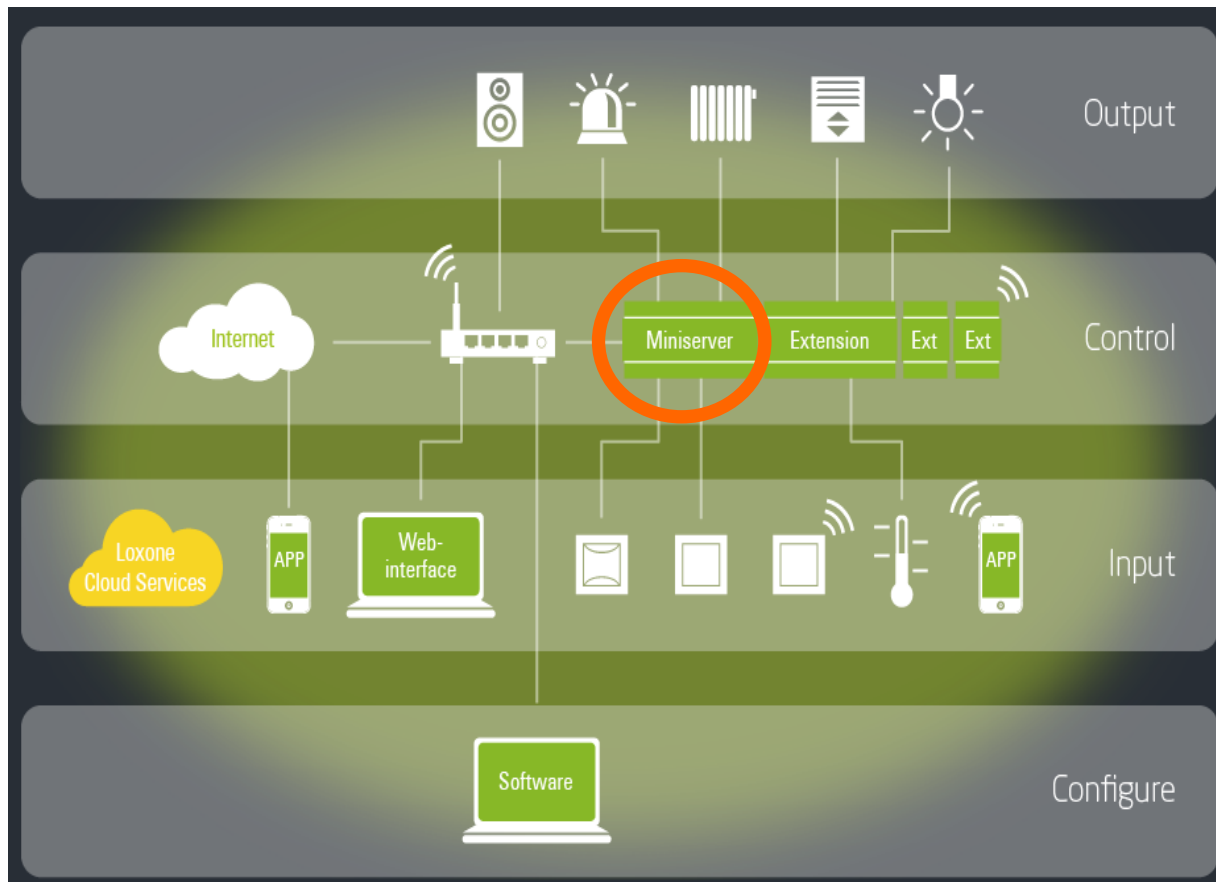


Integration of Field Communication in DDC Wiring to the DDC



Integration of Field Communication in DDC Wiring to the DDC





Loxone Config - ProjectBelimo

Project Management | My Project | Testing | Diagnostics | Network Periphery | Miniserver

Connect | Load from Miniserver | Save in Miniserver | Start LiveView | Search (F5) | Auto Configuration | Device Status | Project Validation | Edit Objects of the Same Type | Add Miniserver | Extension Search | Add Extension | Loxone Link Diagnostics | Link Monitor | Create Backup | Configure Device | Manage Firmware Packages... | Miniserver Firmware Update | Replace Miniserver | Download Statistics

Not Connected | EPIV_V4_MP_Komp.backup* | Clean up project

Properties (Miniserver)

General

Name: Miniserver

Description: Edit...

Hint Text: Edit...

Connection: Miniserver

Display Diagnostic Inputs

Object Type: Miniserver

Settings

Serial Number: 50:4F:94:A0:E4:49

Device type: Miniserver

Local Address: 192.168.0.44

Hostname: MSE449

External Address: dns.loxonecloud.com

External Port HTTP: 80

External Port HTTPS: 443

Miniserver Configuration: Edit...

Installation Location

Installation Location

Panel: Panel

Row in Panel: 1

Position in Cabinet Row: 1

Periphery

Category: All | Boom: All

Preset filters: No Filters

☒ Filter automatically

- Users and Permissions
 - C Constants
 - M Memory Flags
 - (x) System Variables
 - Operating Modes
 - Operating Times
 - Time Functions
 - Remote Controls
 - Push Notifications: Registered devices
- Weather Service
- Weather Station
- Trunks
- Miniserver (Panel R01 P01) (~3% Utilization)
 - Digital Inputs
 - Analogue Inputs
 - Digital Outputs
 - Virtual Inputs
 - Virtual Outputs
 - Intercom
 - Network Periphery
 - Network Intercommunication
 - Modbus Server 1
 - Belimo Thermal Energy Meter Modbus TCP
 - Actuators
 - Sensors
 - Messages
 - Lighting Groups
 - SD Card
 - Link (1/30 Devices)
 - Modbus Extension (Panel R01 P02) (1/32 Devices) (29/253 Sensors)
 - Belimo EPIV 4 Modbus RTU
 - Tree (1/50 Devices)

Miniserver

Central control unit for all kinds of automation tasks.

[More information](#)

ProjectBelimo x

Power Supply: +24V, GND

Link: BU, WH

Tree: GN, WH

LAN

Miniserver

SN: 50:4F:94:A0:E4:49

Q1 - Actuator (Relay) 1 (Not Assigned, Not Assigned)

Q2 - Actuator (Relay) 2 (Not Assigned, Not Assigned)

Q3 - Actuator (Relay) 3 (Not Assigned, Not Assigned)

Q4 - Actuator (Relay) 4 (Not Assigned, Not Assigned)

Q5 - Actuator (Relay) 5 (Not Assigned, Not Assigned)

Q6 - Actuator (Relay) 6 (Not Assigned, Not Assigned)

Q7 - Actuator (Relay) 7 (Not Assigned, Not Assigned)

Q8 - Actuator (Relay) 8 (Not Assigned, Not Assigned)

Input 1 (Not Assigned, Not Assigned) - I1

Input 2 (Not Assigned, Not Assigned) - I2

Input 3 (Not Assigned, Not Assigned) - I3

Input 4 (Not Assigned, Not Assigned) - I4

Input 5 (Not Assigned, Not Assigned) - I5

Input 6 (Not Assigned, Not Assigned) - I6

Input 7 (Not Assigned, Not Assigned) - I7

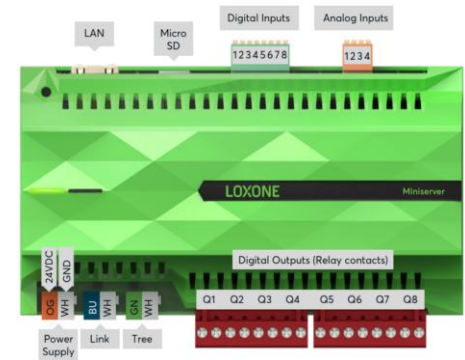
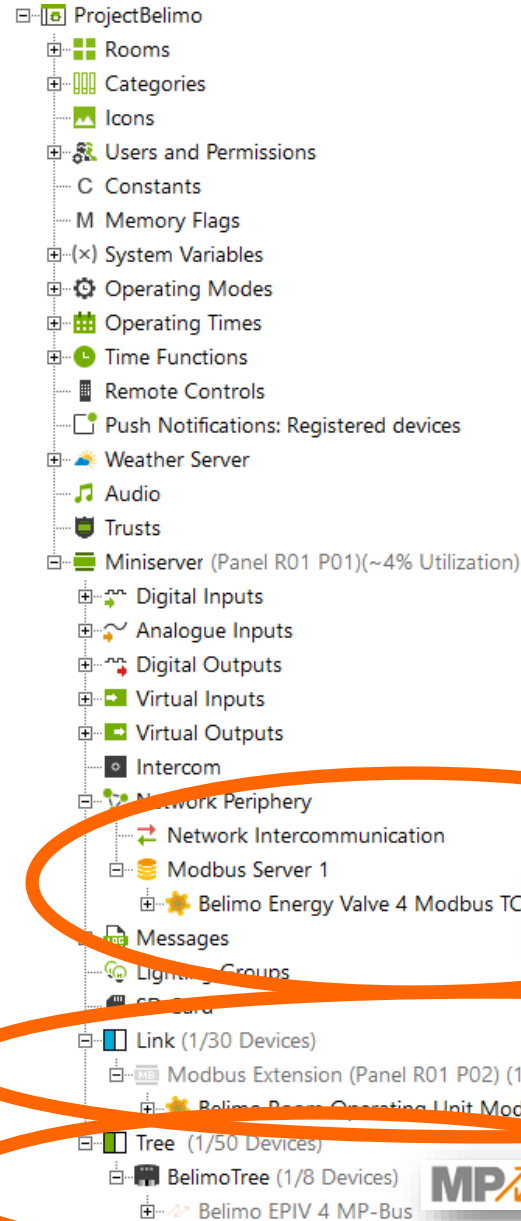
Input 8 (Not Assigned, Not Assigned) - I8

AI1 - Voltage 1 (Not Assigned, Not Assigned) - AI1

AI2 - Voltage 2 (Not Assigned, Not Assigned) - AI2

AI3 - Voltage 3 (Not Assigned, Not Assigned) - AI3

AI4 - Voltage 4 (Not Assigned, Not Assigned) - AI4



Loxone: Example TEM Modbus TCP

General

Name	Absolute Power Cooling in sele...
Description	
Hint Text	Edit...
Connection	AMQ65
Category	Not Assigned
Room	Not Assigned
Object Type	Analogue Sensor

User Interface

☐ Use in User Interface

Rating: ★★★★★★★★★★

Statistics

☐ Statistics

Permissions

Authorised Users / Groups: Edit...

Settings

☐ Display Error Output

IO Address: 31

Command: 4 - Read Input Register (3x)

Data Type: 32-bit Unsigned Integer

☒ 16-bit register

☐ Order of Registers

☐ Byte Order

Polling cycle [s]: 5

Correction

Input Value 1	0
Target Value 1	0
Input Value 2	100
Target Value 2	10

Logging/Mail/Call/Track

Validation

☐ Use Validation

☒ Monitor validation

Receive timeout: 25

Default Value: 0

Display

Analogue Sensor [kW]

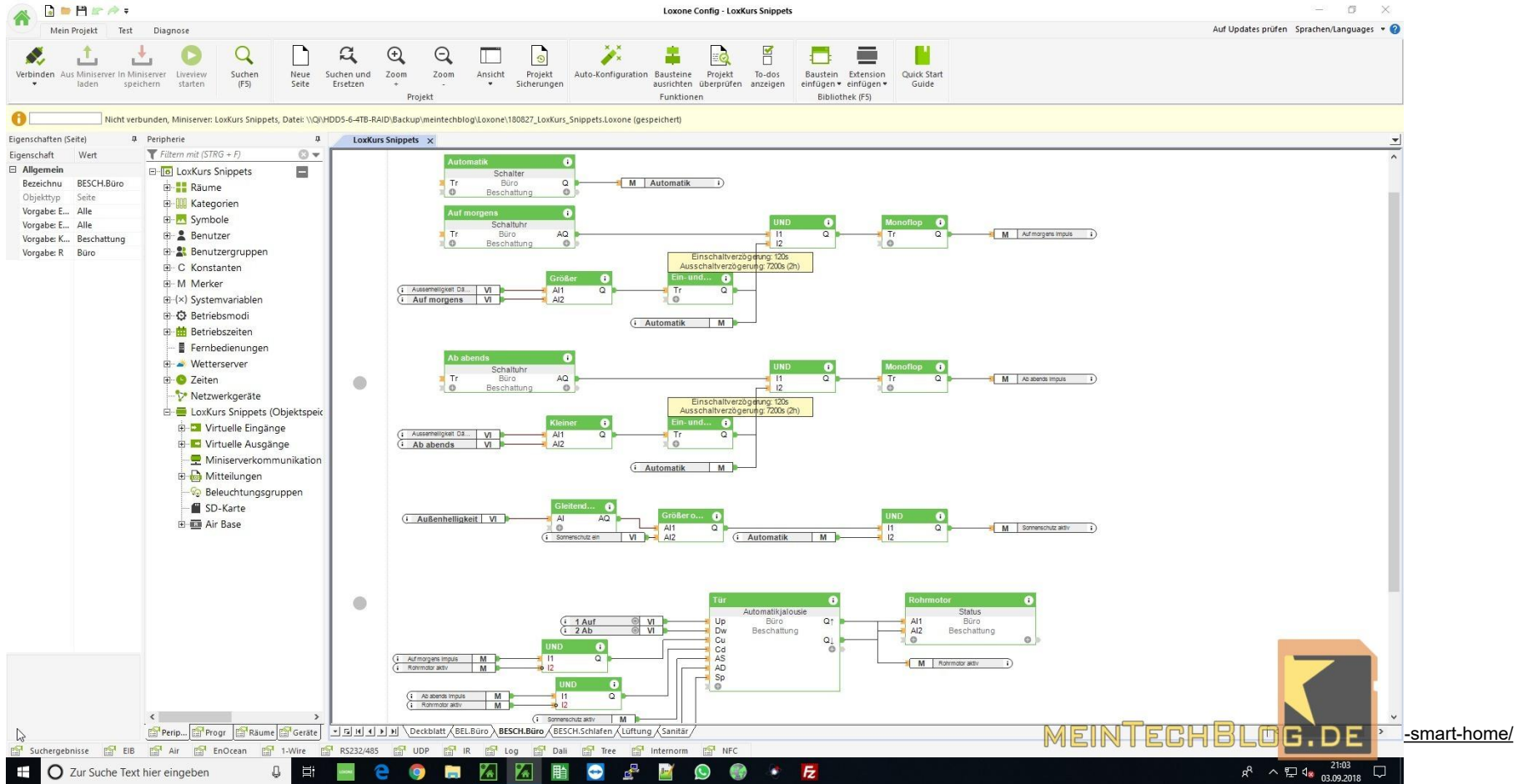
Category: All **Room:** All

Preset filters: No Filters

☒ Filter automatically

Network Periphery

- Network Intercommunication
- Modbus Server 1
 - Belimo Thermal Energy Meter Modbus TCP
 - Actuators
 - Communication Watchdog (AQ)
 - Select Meter Register (AQ)
 - Sensor 1 Passive Sensor Type (AQ)
 - Sensor 1 Type (AQ)
 - Unit Selection Energy (AQ)
 - Unit Selection Flow (AQ)
 - Unit Selection Power (AQ)
 - Unit Selection Volume (AQ)
 - Sensors
 - Absolute Power Cooling in kBtu per hour (AI)
 - Absolute Power Cooling in kW (AI)
 - Absolute Power Cooling in selected unit (AI)**
 - Absolute Power Heating in kBtu per hour (AI)
 - Absolute Power Heating in kW (AI)
 - Absolute Power Heating in selected unit (AI)
 - Absolute volumetric flow in gpm (AI)
 - Absolute volumetric flow in liter per second (AI)
 - Absolute volumetric flow in selected unit (AI)
 - Bus termination (AI)
 - DeltaT_F (AI)
 - DeltaT_K (AI)
 - Device type (AI)
 - Energy Cooling in kBtu (AI)
 - Energy Cooling in kWh (AI)
 - Energy Cooling in selected unit (AI)
 - Energy Heating in kBtu (AI)
 - Energy Heating in kWh (AI)

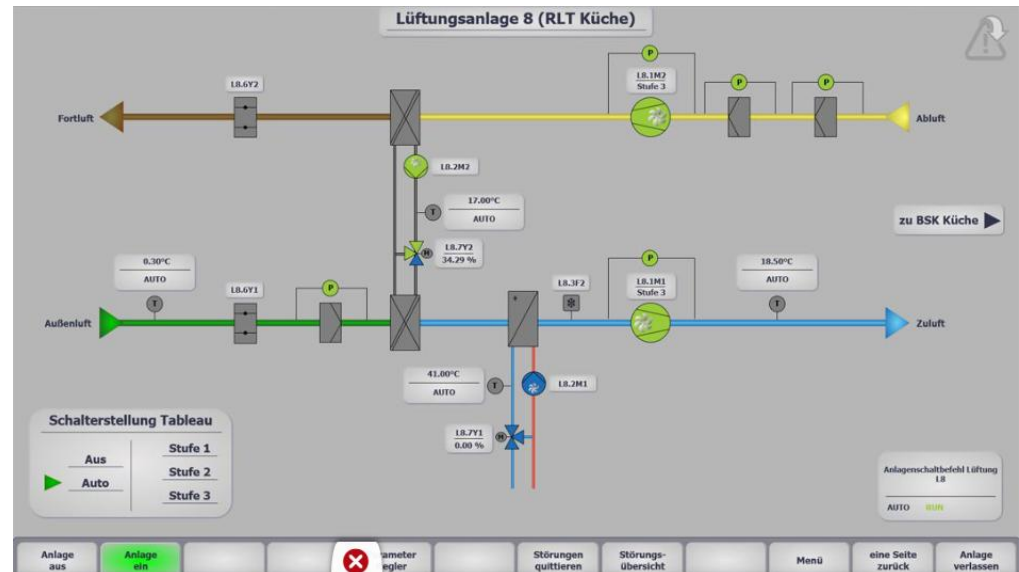


The screenshot displays the Loxone Config software interface. The top menu bar includes options like 'Mein Projekt', 'Test', 'Diagnose', and 'Auf Updates prüfen'. The left sidebar shows a project tree with categories like 'Räume', 'Kategorien', 'Symbole', 'Benutzer', 'Konstanten', 'Merker', 'Systemvariablen', 'Betriebsmodi', 'Betriebszeiten', 'Fernbedienungen', 'Wetterserver', 'Zeiten', 'Netzwerkgeräte', and 'LoxKurs Snippets (Objektspeic)'. The main workspace shows a logic diagram with several components: 'Automatik', 'Auf morgens', 'Ab abends', 'Großer', 'Kleiner', 'Gleitend...', 'Tür', and 'Rohrmotor'. The diagram is connected by lines representing logic flow. A status bar at the bottom shows search results and system information.

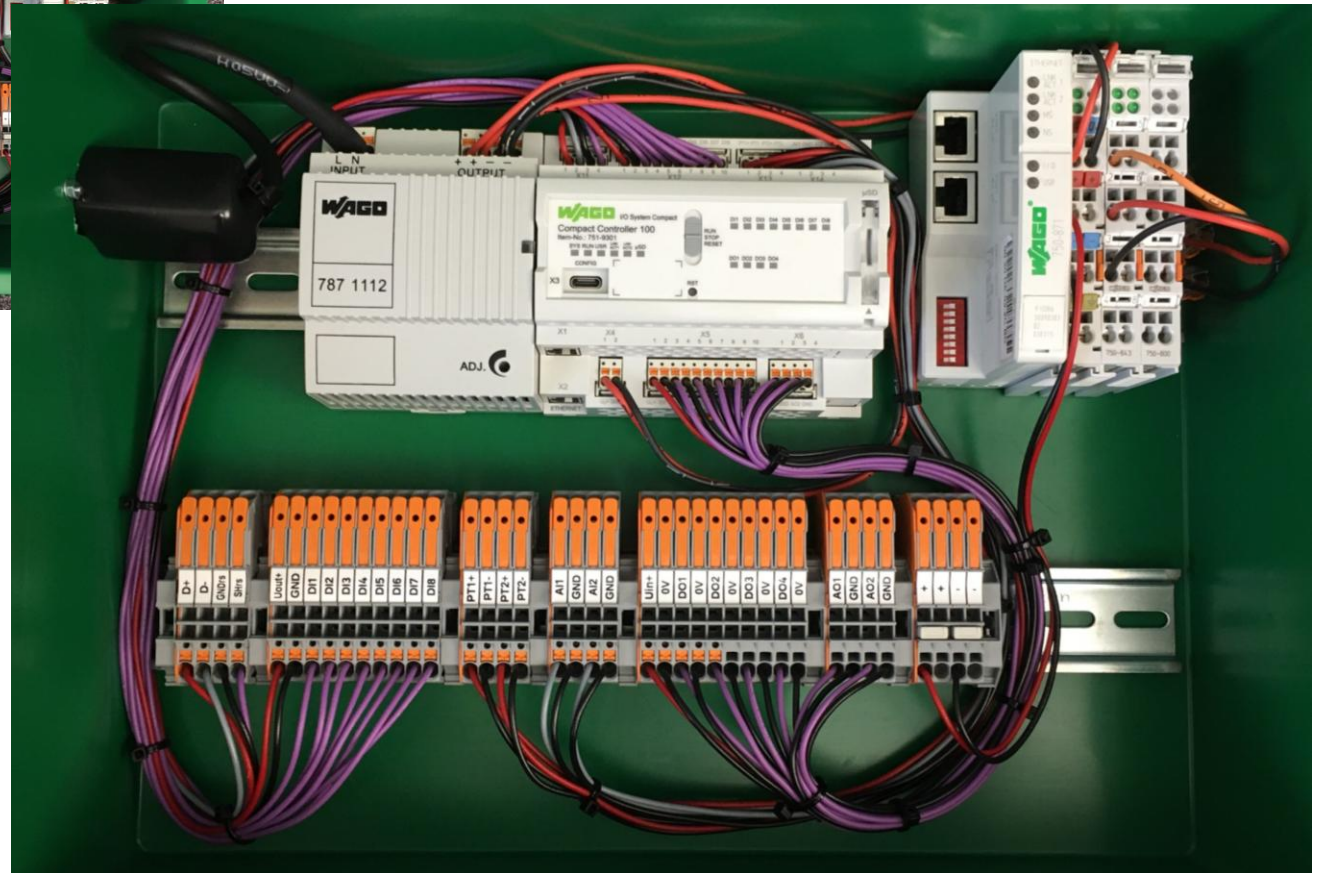
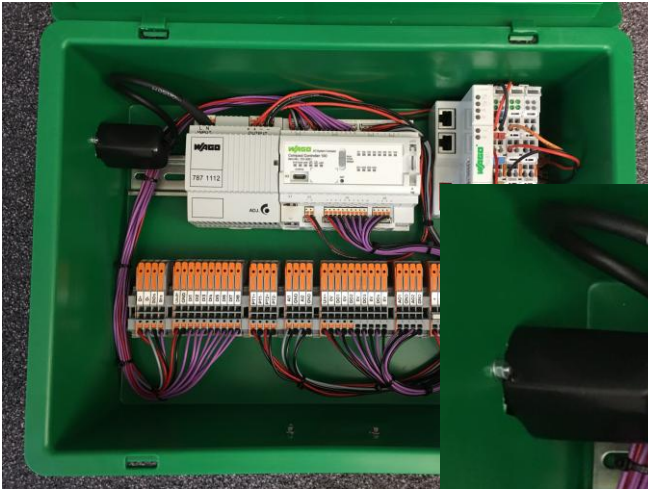
Example: Devices in WAGO



- Supported Protocols by WAGO



WAGO Compact Controller (test box)



WAGO MP-Master - Modul

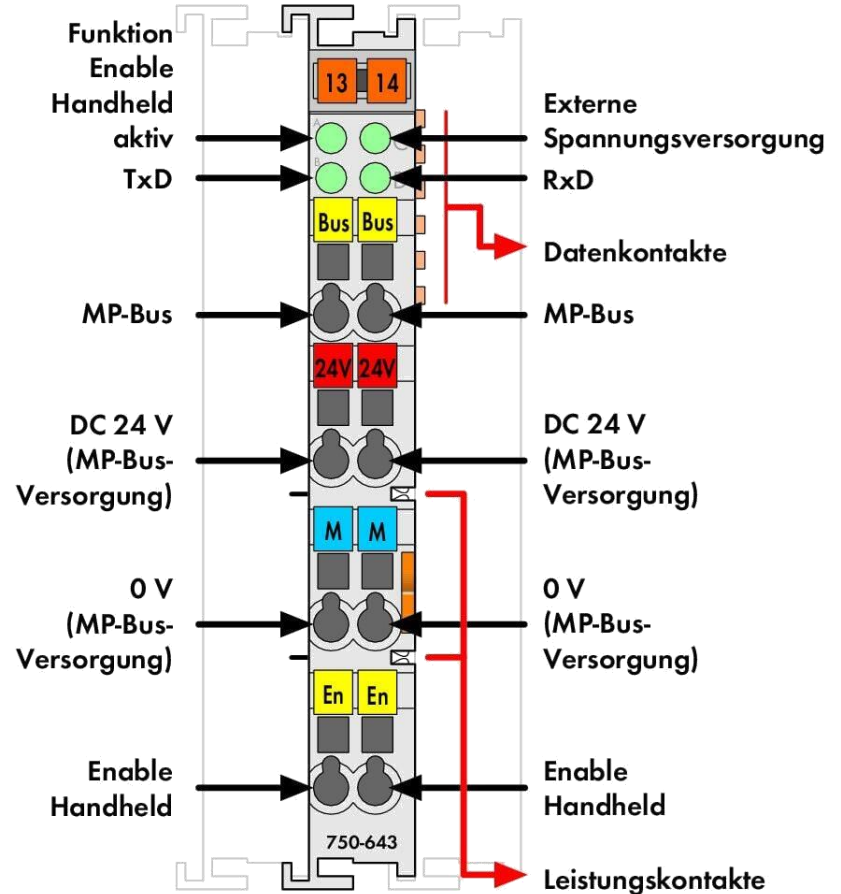


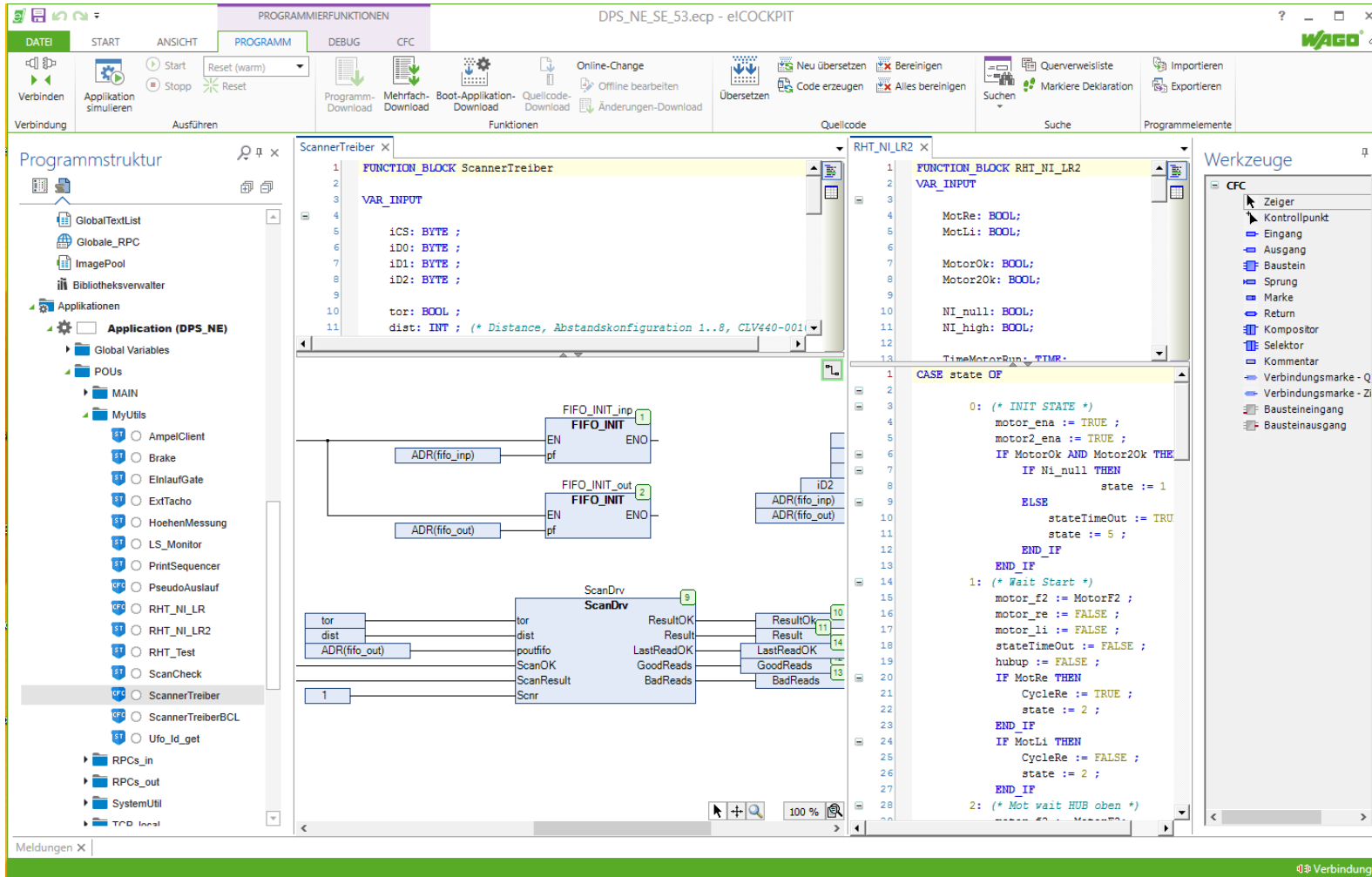
PRODUKTE

LÖSUNGEN

WAGO I/O System 750/753 > Art.-Nr. 750-643

MP-Bus-Master





The screenshot displays the WAGO Engineering software interface for the project "DPS_NE_SE_53.ecp - e!COCKPIT". The interface is divided into several panes:

- Top Bar:** Contains tabs for DATEI, START, ANSICHT, PROGRAMM, DEBUG, and CFC. The PROGRAMM tab is active, showing a toolbar with various functions like Verbinden, Applikation simulieren, Start, Stop, Reset, and Download.
- Left Pane (Programmstruktur):** Shows the project hierarchy, including Global Variables, POU, and MyUtils. The "ScannerTreiber" block is selected under the "MyUtils" folder.
- Center Pane (Ladder Logic):** Displays the ladder logic for the "ScannerTreiber" block. It shows a function block "FIFO_INIT" with inputs "EN" and "pf", and outputs "ENO". The logic includes a "ScanDrv" block with inputs "tor", "dist", "ADR(fifo_out)", and "ScanOK", and outputs "ResultOK", "Result", "LastReadOK", "GoodReads", "ScanResult", and "BadReads".
- Right Pane (Code Editor):** Shows the source code for the "RHT_NI_LR2" function block. The code includes variable declarations for "VAR_INPUT" and "VAR_OUTPUT", and a "CASE state OF" statement with two cases: "0: (* INIT STATE *)" and "1: (* Wait Start *)".
- Bottom Pane (Werkzeuge):** Contains a list of tools and components, including "Zeiger", "Kontrollpunkt", "Eingang", "Ausgang", "Baustein", "Sprung", "Marke", "Return", "Kompositor", "Selektor", "Kommentar", "Verbindungsmaße", "Verbindungsmaße - Zi", "Bausteineingang", and "Bausteinausgang".

Example: MP-Bus Devices in BECKHOFF



The image displays two software interfaces. The top interface is the Beckhoff Information System (BIS) website, showing the 'MP-CMV' function block in the 'TwinCAT 3 PLC Lib: Tc2_MPBus' library. The bottom interface is Microsoft Visual Studio, showing the 'HuaigLibrary' project with the 'pFB_LR24A' function block in the 'pFB_LR24A' file. The 'pFB_LR24A' file is highlighted in the Project Explorer, and the 'pFB_LR24A' function block is highlighted in the Solution Explorer. The 'pFB_LR24A' function block is highlighted in the Solution Explorer, and the 'pFB_LR24A' function block is highlighted in the Solution Explorer.

Beckhoff Information System

Beckhoff Information System

[Select language] Home Kontakt www.beckhoff.de per E-Mail senden Suchen

TwinCAT 3 PLC Lib: Tc2_MPBus

MP-CMV

MP-CMV

MP_Address : bBus
bStart : bError
bSet : iErrorID
bOpen : ActValue
bClose : bMP_Sensor_Digi
bReset : iMP_Sensor_Analog
iSensorType : AirVolume_m3h
SetPoint : iVnom_m3h
MaxVol : act_MaxVol
MinVol : act_MinVol
strDataKL6771 : bErr_ActHunt
TMPolling : bErr_MecTrv
bErr_MecOvld
rTemperature

Dieser Funktionsbaustein dient zur Steuerung und Überwachung eines Volumens. Über MP_Address wird vorgegeben, mit welchem MP-Bus Teilnehmer der Bausatz kommuniziert. Über bStart wird angezeigt, dass der Teilnehmer in der Kommunikation mit dem Antrieb liegt. Die Zeit ist in der Kommunikation mit dem Antrieb angegeben. Die Art des Fehlers ist in der Kommunikation mit dem Antrieb angegeben.

VAR_INPUT

MP_Address : USINT := 1;
bStart : BOOL;
bSet : BOOL;
bOpen : BOOL;
bClose : BOOL;
bReset : BOOL;
iSensorType : INT;
SetPoint : USINT;
MaxVol : WORD;
MinVol : WORD;
strDataKL6771 : DataKL6771;
TMPolling : TIME := t#10S;

MP_Address: MP-Bus Adresse des Slaves.

bStart: Positive Flanke startet den Baustein. Liegt dauerhaft ein TRUE an, so wird zyklisch aktiv.

bSet: Positive Flanke schreibt die Daten MaxVol und MinVol zum Antrieb.

bOpen: Positive Flanke öffnet die Klappen des Antriebs, eine negative Flanke schließt sie.

bClose: Positive Flanke schließt die Klappen des Antriebs, eine negative Flanke öffnet sie.

bReset: Positive Flanke setzt die Fehlermeldungen des Antriebs zurück.

iSensorType: 0: kein Sensor angeschlossen, 1: digitaler Sensor angeschlossen (V), 3: 6: Ausgabe eines Widerstandes in Ohm (3...5 gilt für PT1000, NI1000 und in eine Temperatur sind die entsprechenden Konvertierungsfunktionen zu verwenden).

SetPoint: Soll-Wert des Antriebs (0...1000).

Visual Studio

HuaigLibrary - Microsoft Visual Studio (Administrator)

Dati Bearbeiten Ansicht Projekt Erstellen Debuggen TwinCAT TwinSAFE PLC Team Extras Test Scope Analysieren Fenster Hilfe

Release TwinCAT RT (x64) H10_G01

Projektmappen-Explorer durchsuchen (Strg+U)

Global Variables
POUs
AnlagenSpezifisch
Anlagensteuerung
Grundfunktionen
Hauptprogramme
Kommunikation
Logik
Version
Visualizations
HustechLibrary.tmc
Standard (Standard)
HustechLibrary.Instance

E/A
Geräte
Zuordnungen
HuaigLibTest (nicht verfügbar)
HuaigModbus (nicht verfügbar)
HuaigModbusTcp
SYSTEM
SPS
HuaigModbusTcp
HuaigModbusTcp Project
Externe Datentypen
References
Data types
Global Variables
POUs
Hauptprogramme
Modbus
Basis Klassen
ModClient
pFB_LR24A (FB)
Geräte
pFB_LR24A (FB)
M_Call
M_LeseDatenVerarbeitung
M_SchreibeDatenVerarbeitung
pFB_ModTcpGeneral (FB)
pFB_PRKCA (FB)

pFB_LR24A

eMail : gassers@hustech.ch
Version : 1.0.0.0

Changes :
11.04.2019 TG Initial version

FUNCTION_BLOCK pFB_LR24A EXTENDS pFB_BelimoLibMaster

(* Ausgangs Variablen *)

VAR_OUTPUT
bOut_Geschlossen : BOOL;
bOut_Offten : BOOL;

END VAR

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28

Fehlerliste
Gesamte Projektmappe 0 Fehler 0 Warnungen 0 Mitteilungen
Querweissliste Fehlerliste Ausgabe Suchergebnisse 1 Haltepunkte Ausnahmeseinstellungen

Experience Center – DDC



Questions?





BELIMO®
